50 Special Question For Half yearly exam

(1) $f(x) = \sqrt{9-x^2}$ find domain & Range

(2) Draw graph of following function

a) motulus function

b) greatest integer function

c) signum function

(3) $f(x) = \dfrac{x^2}{1+x^2}$ find Range

(4) find

a) $\sin 15^\circ$

b) $\tan 75^\circ$

c) $\tan 22.5^\circ$

d) $\cot 15^\circ$

(5) To prove

$$\cos 4x = 1 - 8\sin^2 x \cos^2 x$$

(6) To prove.

$$\tan 4x = \frac{4\tan x\,(1-\tan^2 x)}{1 - 6\tan^2 x + \tan^4 x}$$

(7) Find number of words with or without meaning which can be made using all the letter of word AGAIN. If these words are written in a dictionary. what will be the 50th word

(8) How many numbers greater than 1000000 can be formed by using the digit 1, 2, 0, 2, 4, 2, 4 ?

(9) In how many ways can the letters of word PERMUTATIONS be arranged

(i) word start with P and end with S

(ii) Vowels are all together

(iii) There are always 4 letter between P and S

(10) Find number of different 8-letter arrangement can be form the letter DAUGHTER so that

(i) All vowels accur together

(ii) all vowels to not accur together

(11) If α and β are different complex numbers with $|\beta| = 1$ then find

$$\left| \frac{\beta - \alpha}{1 - \bar{\alpha}\beta} \right|$$

(12) If $\left(\frac{1+i}{1-i}\right)^m = 1$ then find least value of m

(13) If $x - iy = \sqrt{\frac{a-ib}{c-id}}$ prove $(x^2+y^2)^2 = \frac{a^2+b^2}{c^2+d^2}$

(14) find conjugate of $\frac{(3-2i)(2+3i)}{(1+2i)(2-i)}$

(15) a) $\left[i^{18} + \left(\frac{1}{i}\right)^{25} \right]^3$ find in $a+ib$

b) $\frac{5+\sqrt{2}i}{1-\sqrt{2}i}$

c) $\frac{(3+i\sqrt{5})(3-i\sqrt{5})}{(\sqrt{3}+\sqrt{2}i)-(\sqrt{3}-i\sqrt{2})}$

(16) find n $\frac{{}^nP_4}{{}^{n-1}P_4} = \frac{5}{3}$

(17) <u>To prove</u> $\cos^2 x + \cos^2\left(x + \frac{\pi}{3}\right) + \cos^2\left(x - \frac{\pi}{3}\right) = \frac{3}{2}$

(18) To prove $\sin 3x + \sin 2x - \sin x = 4 \sin x \cdot \cos\frac{x}{2} \cos\frac{3x}{2}$

(19) find $\sin\frac{x}{2}$, $\cos\frac{x}{2}$, $\tan\frac{x}{2}$

$\tan x = \frac{3}{4}$ $\quad \pi < x < \frac{3\pi}{2}$

(20) To prove

$\cot x \cot 2x - \cot 2x \cot 3x - \cot 3x \cot x = 1$

(21) Prove

$\cos 2x \cos\frac{x}{2} - \cos 3x \cos\frac{9x}{2} = \sin 5x \cdot \sin\frac{5x}{2}$

(22) Find all pairs of consecutive odd natural number, both of which are larger than 10, such that their sum is less than 40

(23) find Range of each of

$f(x) = 2-3x$

$f(x) = x^2+2$

(24) How many words with or without meaning each of 3 vowels and 2 consonants can be formed from letters of $\boxed{\text{INVOLUTE}}$

(25) In how many ways letter of word $\boxed{\text{ASSASSINATION}}$ be arranged so that all are $\boxed{S}$ together.